

Does string theory justify semiclassical gravity at low curvature? Lessons from the black hole information problem

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Abstract

String theory is often cited as the strongest support for treating general relativity (GR), with quantum fields on curved spacetime, as the effective description of gravity wherever curvature is small. We test this support against the black hole information problem. Coarse-graining the exterior of a black hole gives a thermodynamic system with the Bekenstein-Hawking entropy, so the effective description itself must follow the Page curve, and the Page time comes while the horizon curvature is still small. Small corrections perturbative in the string coupling cannot produce the turnaround. The main string-theoretic responses (string spreading, fuzzballs, islands and replica wormholes) all depart from semiclassical gravity at the horizon, violating the conditions of the worldsheet derivation of Einstein's equations. The derivation only establishes that the local dynamics of general relativity holds when local phenomena can truly be considered without considering global effects, suggesting that the string-theoretic support for semiclassical gravity is weaker than often thought. This opens up the possibility of avoiding the circular effective description paradox in philosophy of physics, in which effective gravitational descriptions require global spacetime structure in advance, although they are supposed to provide an effective theory of such structure.

1 Introduction

It is widely held that general relativity (GR) is an effective description of gravitating systems wherever curvature is small, and that quantum fields propagating on general-relativistic space-time solutions are described by effective field theory (EFT) in the same small-curvature regime [1, 2, 3, 4]. Together these make up semiclassical gravity [5, 6]. The black hole information problem challenges semiclassical gravity, since it is widely believed that the conventional view must break down at low curvature, long before the black hole has fully evaporated [7, 8, 9]. The breakdown need not be a complete failure of semiclassical gravity, but the challenge is serious.

Schneider (2024) [4] argues, correctly in our view, that the information problem is one instance of a more general problem: GR as an effective description is not self-sufficient, because it works only once the global spacetime structure has been fixed. If the spacetime contains a high-energy or high-curvature sector, the low-energy regime depends on how that sector is completed, since gauge-invariant observables in GR cannot be local [10, 11]. We call this the *effective description paradox*. The paradox can be avoided if semiclassical gravity is not the full low-energy, low-curvature effective description but a special case of a more general, self-sufficient effective gravitational description. We examine this possibility using hints from the mainstream approaches to the black hole information problem.

String theory is often taken to give the strongest support for the conventional view that semiclassical gravity governs the low-curvature regime, so we start there. Worldsheet conformal invariance requires the target-space metric to satisfy Einstein's equations, up to corrections that are small when spacetime curvature is small compared with the string scale [12, 13], and Huggett and Vistarini [14] argue that the metric is then a coherent state of gravitons, so that GR is the low-energy dynamics of string excitations. Huggett and Wüthrich [15] describe GR as an effective, phenomenal theory of the collective dynamics of strings. On this picture, string perturbation theory around a background spacetime reduces at low energies to quantum fields on that background. The validity conditions are local: curvature small in string units and a small string coupling.

In the conventional view, the horizon of a large black hole is smooth and has small curvature (fuzzballs [16] are the exception), so the string-theoretic support for GR as an effective description seems to apply. Yet most contemporary string-theoretic responses to the information problem imply that semiclassical gravity is not an effective description of an old, evaporating black hole, even at its horizon. This is nearly unavoidable if the Page curve is correct [7, 9]. We therefore need to see where the conventional string-theoretic derivation of GR stops applying.

Our argument proceeds in three steps. First, we argue that the information problem applies to both the effective description and the fine-grained theory. Coarse-graining the black hole exterior of the fine-grained theory gives a thermodynamic system with the Bekenstein-Hawking entropy. This binds the effective description to the Page curve, and the Page time, when the entanglement entropy turns around, comes while the horizon curvature is still small. By

Mathur’s small corrections theorem [17], small corrections perturbative in the string coupling cannot change this. Non-perturbative corrections can change it only if the no-drama equivalence principle, the factorization of the Hilbert space into interior and exterior, or the unitarity of the effective dynamics is given up. Second, we show that each string-theoretic response gives up one of these. Third, we identify what the worldsheet derivation of GR establishes and why it does not apply to the information problem. The derivation justifies local field equations, perturbatively in the string coupling, at fixed target-space topology, for scattering observables. The information problem involves non-perturbative effects and intermediate-time, non-scattering questions, so the derivation is not directly applicable. This opens up a way to avoid the effective description paradox, although the details remain unclear.

2 The conventional ‘effective gravitational description as GR’ view

2.1 Effective descriptions

Following Schneider [4], we use ‘effective description’ as an explanation that captures everything relevant about the system dynamics in a coarse-grained regime, typically the low-energy, low-curvature regime with scale cutoffs. An effective description should be self-sufficient: it should describe the system in question wherever the system stays in the applicable regime. We expect such a description to be an *effective theory*. The term *effective field theory* (EFT) is reserved for an effective theory that uses the field-theoretic formalism.

2.2 Conventional semiclassical gravity view

In the conventional view, GR gives effective descriptions for the low-curvature low-energy regime. For example, GR can be stated as providing the mean-field sector of low-energy quantum gravity [2, 3]. Or, GR can be recast as a spin-2 field theory on a fixed background [18, 19]. Schneider (2024) observes that both can be stated only after the global structure of spacetime has been fixed, since the field observables of GR are defined only once the diffeomorphism class and more of the global structure are given [20]. Any relativist will recognize the point: it makes no sense to talk of the electric field at a particular spacetime location x in general relativity, because x cannot be specified in a gauge-invariant way [10, 11].

The mean-field description of GR restricts attention to the low-curvature, low-energy regime, but GR still needs the global spacetime structure to work. Unless the spacetime forbids the low-curvature, low-energy regime altogether, using GR as an effective theory depends on a completion outside the regime of the effective theory. Therefore, GR is not self-sufficient within the effective-theory regime. The spin-2 formulation presupposes a fixed background, so the global spacetime structure is given in advance and the description is again not self-sufficient.

2.3 Why the effective description matters when a fine-grained description is available

One could respond that this lack of self-sufficiency does not matter: where semiclassical gravity breaks down (for example, near an old black hole), we should simply use the better fine-grained description. This does not settle the matter, for two reasons. First, measurements are made with effective observables, not quantum gravity observables. Furthermore, if semiclassical gravity turns out to be non-unitary, this conflicts severely with the effective-theory unitarity assumption in quantum computation. In the case of black holes, it matters whether an effective description keeps no drama at the horizon, factorization of the Hilbert space into interior and exterior, and unitarity. Second, the models in which the fine-grained description is stated are mostly holographic gravity models, and such a model can produce an effective bulk that is inconsistent with real black holes, which limits its value even as a toy model.

3 General relativity from string theory

The derivation of GR from string theory goes back to work on sigma models and strings in background fields in the 1980s [12, 21, 22, 13]. Since Huggett and Vistarini [14] give the cleanest summary, we largely follow their presentation. Start from the string sigma model with a general target-space metric,

$$S_\sigma = -\frac{1}{4\pi\alpha'} \int d^2\sigma \sqrt{-\gamma} \gamma^{ab} G_{\mu\nu}(X) \partial_a X^\mu \partial_b X^\nu. \quad (1)$$

From the worldsheet point of view, S_σ is a two-dimensional quantum field theory in which the spacetime metric $G_{\mu\nu}(X)$ plays the role of a coupling constant, or rather an infinite set of them, one for each point of spacetime. As in any interacting quantum field theory, these couplings change with the scale at which the theory is probed, and the renormalization group β -function measures how fast they change. The β -function can be computed as a series in powers of $\alpha' = \ell_s^2$, the square of the string length ℓ_s . Each additional power of α' comes with more derivatives of the metric, so the higher terms are negligible when spacetime is curved only on scales much larger than ℓ_s . The first term of the series is

$$\beta_{\mu\nu}^G = \alpha' R_{\mu\nu} + O(\alpha'^2) \quad (2)$$

[12]. A consistent string theory needs the worldsheet theory to be Weyl invariant, which means that its physics cannot depend on the local scale of the worldsheet. The couplings therefore must not run, so $\beta_{\mu\nu}^G = 0$. Keeping only this first term, the condition says that the spacetime metric is Ricci flat, $R_{\mu\nu} = 0$, which is Einstein's equation in vacuum.

Matter enters through the other massless states of the string. The simplest of these is the

dilaton Φ , a scalar field whose constant part sets the string coupling, $g_s = e^{\Phi_0}$. Huggett and Vistarini extend the argument to a background that includes the dilaton (but no antisymmetric tensor field), where Weyl invariance requires [23]

$$\beta^\Phi = \frac{D-26}{6} - \frac{\alpha'}{2} \nabla^2 \Phi + \alpha' \nabla_\omega \Phi \nabla^\omega \Phi + O(\alpha'^2) = 0, \quad \beta_{\mu\nu}^G = \alpha' (R_{\mu\nu} + 2\nabla_\mu \nabla_\nu \Phi) + O(\alpha'^2) = 0. \quad (3)$$

Here D is the number of target-space dimensions. Weyl invariance can fail through a quantum effect, the Weyl anomaly, and the constant term in β^Φ measures it. Each of the D fields X^μ contributes to the anomaly, and the ghost fields introduced when the worldsheet metric is fixed contribute -26 , so the anomaly cancels only when $D = 26$. This value is the *critical dimension* of the bosonic string. The superstring theories that are candidates for describing our world have critical dimension ten, and their metric and dilaton obey the same first terms of the β -functions, with the constant term vanishing at $D = 10$ instead. In the critical dimension a constant dilaton satisfies $\beta^\Phi = 0$, apart from the higher powers of α' .

The conditions $\beta_{\mu\nu}^G = \beta^\Phi = 0$ lead to the field equations of a spacetime action for the metric and the dilaton. Rescaling the metric by a power of e^Φ gives the ‘Einstein-frame’ metric $\tilde{G}_{\mu\nu}$, in which the action takes the standard Einstein-Hilbert form. In the critical dimension the metric equation then reads

$$\tilde{R}_{\mu\nu} - \frac{1}{2} \tilde{G}_{\mu\nu} \tilde{R} = \frac{4}{D-2} \left(\partial_\mu \Phi \partial_\nu \Phi - \frac{1}{2} \tilde{G}_{\mu\nu} \partial_\omega \Phi \partial^\omega \Phi \right) + O(\alpha') \quad (4)$$

[23]. The right-hand side is the stress-energy tensor of a massless scalar field. The dilaton behaves as ordinary matter, and the curvature of spacetime responds as Einstein’s equations require. Other background fields work the same way. For the heterotic string in a background Yang-Mills field, Weyl invariance requires the Yang-Mills equations to hold, and the metric β -function gains a term proportional to the stress-energy of the gauge field, suppressed by one further power of α' relative to the curvature term [13, 24].

Outside the critical dimension, a constant dilaton no longer solves $\beta^\Phi = 0$. Flat spacetime is still a solution if the dilaton grows linearly in one direction, $\Phi = V_\mu X^\mu$ with $V_\mu V^\mu = \frac{26-D}{6\alpha'}$ [23]. Since $g_s = e^\Phi$, the string coupling then grows exponentially in that direction, and string perturbation theory breaks down there. Huggett and Vistarini use this to argue that Weyl invariance, and with it the field equations above, cannot simply be dropped. Without Weyl invariance, the conformal factor of the worldsheet metric no longer drops out of the theory. It becomes an extra dynamical field that behaves like an additional target-space dimension with a linear dilaton, and perturbation theory breaks down.

Huggett and Vistarini then interpret $G_{\mu\nu}$ and the other background fields as coherent states of string excitations. Just as a classical electromagnetic field can be described as a coherent state of many photons, a curved background amounts to a coherent state of many gravitons: formally, the sigma model on a curved background is the flat-space theory with the graviton

vertex operator exponentiated, which builds a coherent state. The gravitational field is then itself made of string states, and the field equations say which coherent states are consistent backgrounds.

Since the theory does not fix a spacetime in advance, Huggett and Vistarini conclude that string theory is background independent: any solution of the equations is an allowed background.

Two expansions control the derivation. The first is the α' expansion. The $O(\alpha'^2)$ terms in the β -functions are negligible only if the curvature radius of the background is large compared with the string length $\ell_s = \sqrt{\alpha'}$. The second is the genus expansion in the string coupling. A string worldsheet can have any number of handles, and a worldsheet with h handles is weighted by g_s^{2h-2} . The β -functions above come from the short-distance behavior of the sigma model on a worldsheet without handles, so they give the genus-zero contribution. The conditions the derivation needs are local: small curvature in string units and small g_s wherever the strings propagate.

3.1 Constraints required for the derivation

The derivation holds only under four constraints. Where a situation violates one of them, the derivation says nothing about whether GR applies.

(i) *Locality*. The β -functions are local and hold on any target manifold. They fix nothing about topology, causal structure or asymptotics. Picking out a solution requires a separate choice of global structure.

(ii) *Perturbation theory at fixed topology*. The derivation is perturbative in g_s , with the target-space topology held fixed, and the equations above are only the first term of that expansion. Worldsheets with handles produce divergences that are cancelled by shifting the background, so the background equations are themselves corrected order by order in g_s [25]. Effects non-perturbative in g_s , including sums over target-space topologies, do not appear at any order. If such effects matter in the low-energy, low-curvature regime, the derivation does not justify GR there.

(iii) *The scattering regime*. String perturbation theory computes S-matrix elements. Its low-energy agreement with EFT is established for scattering between asymptotic states, typically in regions that embed in asymptotically flat spacetimes and away from horizons. Huggett and Wüthrich's account of the emergence of local spacetime structure [15] is formulated in this setting. As Schneider (2024) [4] notes, this only recovers local rather than global properties of spacetime.

(iv) *Coherent states and expectation values*. The coherent-state reading shows that the expectation value of the metric obeys Einstein's equations. It does not show that the Hilbert space of the EFT on the background is the right state space.

The mainstream approaches to the black hole information problem, discussed below, tend to violate these constraints.

4 The black hole information problem at low curvature

Hawking argued that black hole evaporation turns a pure-state universe into a mixed-state universe [26]. Belot, Earman and Ruetsche [27] give an algebraic quantum field theory argument [28] for the non-problematic nature of this pure-to-mixed evolution: unitarity is guaranteed only for evolution between Cauchy surfaces, and the conventional choice of foliation surfaces is not Cauchy surfaces. Wallace [29] distinguishes this evaporation-time paradox from the Page-time paradox and argues that quantum gravity physicists usually mean the latter, with Raju [30] noting that final-state purity is easy to reconcile with a mixed-looking state when the available observables are simple.

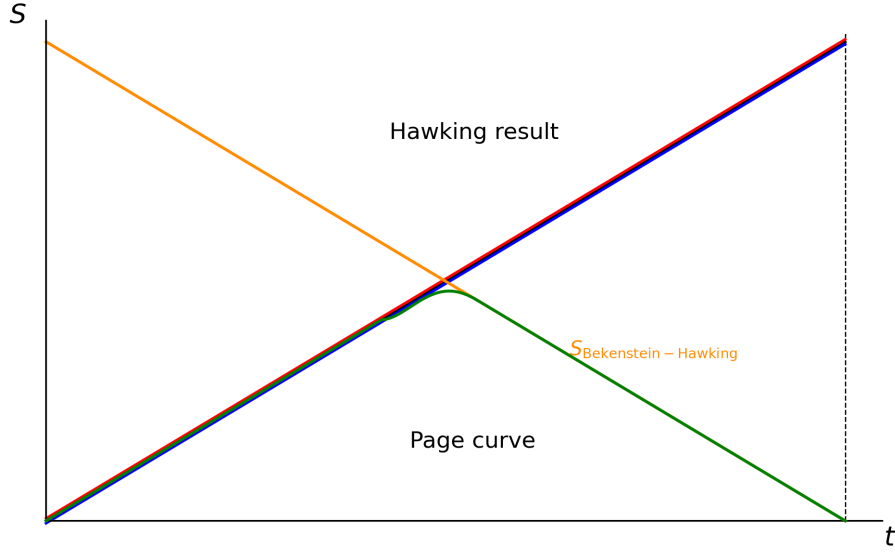


Figure 1: The Bekenstein-Hawking entropy (monotonically decreasing yellow line) is an approximate coarse-grained upper bound to entanglement entropy. The Hawking result (monotonically increasing purple line) is predicted by semiclassical effective field theory, and exceeds the bound after the Page time. The ideal Page curve is the green curve.

Black hole thermodynamics, first derived without quantum physics [31, 32], gives the first law in natural units,

$$dM = \frac{\kappa}{8\pi} dA + \Omega dJ + \Phi dQ. \quad (5)$$

Hawking radiation [33] fixes $T = \kappa/2\pi$, and the thermodynamic entropy is

$$S_{BH} = \frac{A}{4}. \quad (6)$$

This Bekenstein-Hawking entropy should bound the entanglement entropy between the black hole and its radiation, but semiclassical gravity predicts that the entanglement entropy grows monotonically, so the two curves cross near the Page time [7] (Figure 1).

4.1 The bound constrains the effective description

Does the bound apply to the entanglement entropy computed in semiclassical gravity, or only to the fine-grained entanglement entropy? The literature implicitly assumes the latter, and that implicitness is itself a problem. If that were all, semiclassical gravity could still describe everything coarse-grained while only the fine-grained theory changes, and the string-theoretic support for the conventional view would be untouched.

The reason for the trouble is that Bekenstein-Hawking entropy is a classical, coarse-grained thermodynamic entropy. To see this, take the fine-grained theory and coarse-grain the black hole exterior. On the conventional view, the resulting effective description of the exterior is a thermodynamic system with the Bekenstein-Hawking entropy. One could say that semiclassical gravity over-counts degrees of freedom to avoid this issue, but that is exactly the problem: over-counting implies that an effective description derived from the fine-grained theory may not be semiclassical gravity. On this thermodynamic view, large corrections cannot be deferred to late evaporation. The Page curve requires large corrections to semiclassical gravity (or to our understanding of it) around the Page time, long before the black hole has fully evaporated.

An epistemic defense of semiclassical gravity is still available. Ignorance can lead one to represent a pure state as a mixed one, as when one assigns probability $1/2$ to each face of a coin that has landed but has not been looked at, and on epistemic interpretations of quantum mechanics quantum states in general represent an agent's information rather than physical reality [34, 35, 36]. Semiclassical gravity, i.e. quantum field theory in curved spacetime with backreaction [5, 6], can be treated in the same way, as an epistemic quantum theory. Nothing prevents this reading: we can keep the epistemically coarser description as long as we accept that it may surprise us. However, quantum gravity corrections to the effective theory are meant to remove exactly this kind of ignorance, so the defense fails unless we admit large actual corrections to semiclassical gravity.

A second possible defense relies on the non-isometric map from the effective to the fundamental description [37]. This map acts between states. The final effective universe state is pure, but the final effective state of the black hole exterior remains mixed, and the map corrects it to a pure state only in the fine-grained fundamental theory. If the map is instead taken to act between states within an effective theory, it commits to large corrections within that theory. So even if the effective state is wrong, we can correct it to a fine-grained state, but this does not defend semiclassical gravity.

4.2 What kinds of correction are available for Page-curve turnaround?

Two kinds of string-theoretic correction could turn the Page curve around: corrections perturbative in g_s and in α' , and corrections non-perturbative in g_s .

Perturbative corrections. Mathur’s small corrections theorem [17] models Hawking radiation as maximally entangled pairs $|\Psi\rangle_{pair} = \frac{1}{\sqrt{2}}(|0\rangle_c|0\rangle_b + |1\rangle_c|1\rangle_b)$ of outgoing and infalling qubits. If each pair state deviates from $|\Psi\rangle_{pair}$ by an amount ϵ , strong subadditivity implies that the entanglement entropy of the radiation grows by at least $\ln 2 - 2\epsilon$ at every emission, so reversing the growth requires $\epsilon = O(1)$. Corrections that are perturbative in g_s , and in α' at low curvature and low center-of-mass energy, are small in this Mathur-theorem sense. The failure of semiclassical gravity cannot come from small perturbative corrections around the black hole background. Section 5.1 discusses a situation near horizons where string perturbation theory is not in this low-energy regime.

Non-perturbative corrections. Corrections of order $e^{-S_{BH}}$ are non-perturbative in g_s . Newton’s constant scales as $G_N \propto g_s^2$ in string units [23], so a black hole of fixed size in string units has $S_{BH} = A/4G_N \propto 1/g_s^2$. Replica wormholes, discussed in Section 5.3, are the best-known example [38, 39]. In the coherent-state picture of Section 3, the black hole background is a coherent state of gravitons and the Hawking quanta are excitations on top of it. A non-perturbative correction acts on this combined state, and the question is how far it can move the state away from its semiclassical evolution. A generalization of Mathur’s theorem answers this question and shows what such corrections need to give up. Let $U_{eff,t}$ be the unitary evolution of the corrected effective description over a time step Δt , and $U_{EFT,t}$ the semiclassical evolution, which keeps the pair entanglement across the horizon that produces Hawking’s result. Let $|\psi_t\rangle$ be the semiclassically evolved state at step t , and suppose that at each step

$$U_{eff,t}|\psi_t\rangle = \alpha_t U_{EFT,t}|\psi_t\rangle + \sqrt{1 - |\alpha_t|^2} |\phi_t\rangle, \quad |\alpha_t|^2 = 1 - e^{-S(t)}, \quad (7)$$

with $S(t)$ large at every t . Here $\alpha_t = \langle \psi_t | U_{EFT,t}^\dagger U_{eff,t} | \psi_t \rangle$ is taken to be real and non-negative. This is no restriction, because $U_{EFT,t}$ may depend on the state and a phase can be absorbed into it without changing the physical state; without it, a relative phase alone would make $|\alpha_t - 1|$ of order one. Let ϵ_k be the norm of the difference between the U_{eff} -evolved and U_{EFT} -evolved states after k steps. Unitarity of U_{eff} preserves the norm of the deviation carried over from the previous step, so the triangle inequality gives $\epsilon_k \leq |\alpha_k - 1| + \sqrt{1 - |\alpha_k|^2} + \epsilon_{k-1}$. With α_k real and non-negative, $|\alpha_k - 1| = 1 - \alpha_k \leq 1 - \sqrt{1 - e^{-S_k}} \leq e^{-S_k}$ and $\sqrt{1 - |\alpha_k|^2} = e^{-S_k/2}$, so $\epsilon_k \lesssim \epsilon_{k-1} + e^{-S_k/2}$, and induction gives

$$\epsilon_T \lesssim (T + 1) e^{-S_{\min}/2}, \quad S_{\min} \equiv \min_{t \leq T} S(t). \quad (8)$$

We leave $S(t)$ unspecified on purpose. The Bekenstein-Hawking entropy is a natural choice, but the argument only needs $S(t)$ to be large, so it covers any proposed scale of small corrections. $U_{eff,t}$ and $U_{EFT,t}$ may depend on the state and on time, so state-dependent maps do not evade the result unless they also violate unitarity. If $U_{EFT,t}$ is non-unitary, the EFT-evolved state is renormalized to unit norm at each step, as is done in practice, or $U_{EFT,t}$ is taken to be a contraction.

The number of steps needed to reach the Page time grows only polynomially with the entropy (in four dimensions the evaporation time scales as $M^3 \propto S_{BH}^{3/2}$), so the deviation is still exponentially small at the Page time. Non-perturbative corrections can therefore produce the required failure only if at least one assumption goes: the no-drama equivalence principle (the corrections are not small near the horizon), factorization of the Hilbert space into interior and exterior (the entropy question changes its meaning), or unitarity of the effective dynamics. Losing no drama means that semiclassical gravity fails near the horizon despite the low curvature. Losing factorization means that the Hilbert space of semiclassical gravity is not the right state space, and that the division into interior and exterior needs to be reconsidered. Losing unitarity means that the effective description no longer evolves as a quantum theory.

5 String-theoretic responses and the failure of semiclassical gravity

Section 4.2 leaves two ways for the effective description to follow the Page curve. Either the corrections to semiclassical evolution are not small after all, or they are non-perturbative and give up no drama, interior/exterior factorization, or unitarity of the effective dynamics. Each of the string-theoretic responses to the information problem takes one of these routes, except holography of information, which denies that the fine-grained entropy of the radiation follows a Page curve at all (Section 5.3). Every response, including holography of information, violates a different constraint required for the worldsheet derivation (for the ‘constraints’ see Section 3.1).

5.1 Longitudinal string spreading and complementarity

The earliest string-theoretic response comes from perturbative string theory itself. ‘String spreading’ is the standard term in this literature for the growth of a string’s apparent size when it is observed with finer time resolution, which is equivalent to observing it from a frame in which it is highly boosted. ‘Longitudinal’ spreading is the growth along the direction of the boost, and this effect becomes relevant near horizons. Susskind [40] argued that a string’s transverse and longitudinal extent grows as it is probed at higher resolution, so that a string falling toward a horizon appears to an exterior observer to spread over the stretched horizon. Lowe, Polchinski, Susskind, Thorlacius and Uglum [41] found signs in light-cone string field theory that string interactions can be non-local over large distances for highly boosted configurations, which are exactly the configurations that relate infalling and late-time exterior observers near a horizon. Dodelson and Silverstein [42] argued, from string amplitudes, that longitudinal spreading can lead to a breakdown of EFT near black hole horizons for late-time infallers, even though the curvature there is small.

These results do not rely on non-perturbative effects. They exploit a violation of the

scattering regime, constraint (iii) of Section 3.1. The worldsheet derivation's validity criterion concerns $\alpha'R \ll 1$, but the low-energy agreement between string scattering and EFT also requires the scattering itself to be at low center-of-mass energy, $\alpha's \ll 1$ with s the squared center-of-mass energy. Near the horizon of a Schwarzschild black hole of mass M , the relative boost between two quanta that fall in at times separated by Δt (as measured at infinity) grows like $e^{\kappa\Delta t}$, with surface gravity $\kappa = 1/4M$, so that

$$s \sim E_1 E_2 e^{\kappa\Delta t} \quad (9)$$

for asymptotic energies E_1 and E_2 [43, 42]. The condition $\alpha's \ll 1$ fails once $\Delta t \gtrsim \kappa^{-1} \ln(1/\alpha'E_1 E_2)$. For quanta with energies of order the Hawking temperature this is of order $M \ln(M/\ell_s)$, a time logarithmic in the mass and far shorter than the Page time, which scales as M^3 . The curvature at the horizon, $R \sim M^{-2}$ for the components of the Riemann tensor (the Ricci tensor of the Schwarzschild solution vanishes), stays small in string units during this whole period, since $\alpha'R \sim \ell_s^2/M^2 \ll 1$. A late infaller near the horizon satisfies the curvature condition of the worldsheet derivation, but not the energy condition that decides whether string amplitudes agree with EFT.

String spreading escapes Mathur's small corrections theorem because the theorem assumes that each new pair is created near the horizon in a state within $\epsilon \ll 1$ of the vacuum pair state, independently of what fell in earlier. This is the locality and no-drama content of the near-horizon EFT. If interactions between late quanta and earlier infalling matter occur at $\alpha's \gtrsim 1$, the stringy amplitudes differ from their EFT counterparts at order one, and the near-horizon evolution can depend on earlier matter in a way the EFT does not allow. The deviation ϵ is then not small, even though the effect is perturbative in g_s . What breaks down is the α' expansion of the amplitudes, while the g_s expansion remains under control.

The idea has since moved to the margins of research on the information problem, although it is still pursued [44]. There are four main reasons. First, it has not explicitly been shown that these effects modify the outgoing Hawking pairs at order one, rather than only what late infallers encounter, and the proposal has not explicitly produced a Page curve. Second, its strongest evidence comes from light-cone gauge estimates. Dodelson and Silverstein also found comparable longitudinal nonlocality in gauge-invariant string scattering amplitudes [45, 46], but these amplitudes are computed in flat spacetime. The scattering kinematics are chosen to imitate an early and a late infaller, and no amplitude has been computed on an actual black hole background, so applying the result at a horizon remains an extrapolation [42]. Third, string spreading was proposed in support of black hole complementarity, but the firewall argument of Almheiri, Marolf, Polchinski and Sully [8] persuaded many that complementarity in its original form is inconsistent. Fourth, since 2019 the island formula and replica wormholes have reproduced the Page curve from the semiclassical gravitational path integral, without stringy dynamics (Section 5.3), and most work on the information problem has moved there.

5.2 Fuzzballs

The fuzzball proposal [16], a minority position within string theory, replaces the black hole by horizonless microstate geometries whose structure extends to the horizon scale. Semiclassical gravity then fails for the usual reason an EFT fails, because new degrees of freedom become relevant. The difference is that in a large black hole they appear where the curvature is small. The proposal gives up the no-drama equivalence principle.

In terms of Section 3.1, fuzzballs violate constraint (i). Each microstate geometry has a different global structure from the classical black hole, with no event horizon, and the classical black hole geometry is at best an average over them. The worldsheet derivation constrains the local field equations each microstate obeys, but it cannot tell us that the smooth black hole geometry, rather than the ensemble of microstates, is the right effective description.

Schneider [47] refines this point, building on Huggett and Matsubara (2021) [48]: nothing forbids us from taking the effective description of a fuzzball to respect the smooth black hole geometry. If we follow the conventional fuzzball story, in the semiclassical limit relevant to a distant observer, the fauxrizon, which is what Huggett and Matsubara call the surface at which effective spacetime ends in a fuzzball, disappears and only an exterior black hole spacetime remains. However, nothing yet prevents that exterior from being extended through a horizon into an interior, and Schneider argues that the fuzzball proposal does not by itself solve the evaporation paradox, and says it would do so only if the stringy structure at the fauxrizon constrained the global spacetime structure of the effective and semiclassical description - this constraint has not been demonstrated.

5.3 Holography, islands and replica wormholes

The mainstream approaches often rely on AdS/CFT [49], which defines string theory in asymptotically AdS spacetimes non-perturbatively through a dual conformal field theory. The Ryu-Takayanagi formula gives the entanglement entropy of a boundary region A as

$$S(A) = \frac{\text{Area}(\gamma_A)}{4G_N}, \quad (10)$$

where γ_A is the minimal extremal surface in the bulk homologous to A [50]. The islands proposal generalizes this to the radiation R ,

$$S(R) = \min_X \text{ext}_X \left[\frac{\text{Area}(X)}{4G_N} + S_{\text{semicl}}(R \cup I_X) \right], \quad (11)$$

extremizing over quantum extremal surfaces X and including any island region I_X behind the horizon [51, 52, 53]. (The proposal is disputed within holographic gravity. One objection is that islands require a massive graviton [54], which we do not address here.)

On its potentially naive reading, after the Page time the Hilbert space of the radiation includes

part of the interior. This breaks interior/exterior factorization, which implies a failure of the state space of semiclassical gravity. However, the local geometry near the horizon shows no sign of this, and the term S_{semicl} is still computed with quantum fields on a smooth background. Yet the semiclassical description no longer determines which fields belong to the radiation.

Replica wormholes explain where the semiclassical entropy calculation fails [38, 39]. Gluing n copies of the Euclidean path integral cyclically along R gives

$$\text{Tr } \rho_R^n = \frac{Z_n}{Z_1^n}, \quad Z_n = \int_{\partial g = \mathcal{M}_n} \mathcal{D}g \, e^{-I_{grav}[g]}, \quad (12)$$

where Z_n includes every geometry with the replicated boundary $\mathcal{M}_n$, including geometries that connect the replicas. In the ordinary use of the replica trick, Z_n contains only the n copies of the original geometry glued along R , and no such connecting geometries. The Rényi entropies $S_n(R) = \frac{1}{1-n} \ln \text{Tr } \rho_R^n$ then give $S(R) = \lim_{n \rightarrow 1} S_n(R)$ (‘replica trick’), which follows the Page curve [55, 56]. In simple models, the connecting geometries contribute to $\text{Tr } \rho_R^n$ with a weight of order $e^{-(n-1)S_{BH}}$, compared with $e^{-(n-1)S_{rad}}$ for the disconnected geometries, so they dominate once the radiation entropy exceeds S_{BH} , i.e. after the Page time. Their weight is non-perturbative in g_s , and they change the topology of the geometry. Replica wormholes therefore violate several constraints required for the string-theoretic derivation of GR: these contributions are absent from string perturbation theory at fixed target-space topology, and the calculation itself determines which region counts as ‘interior’ for the radiation’s entropy. Whether the Euclidean saddles have a Lorentzian, Hilbert-space counterpart in the effective description is still disputed, and so is whether such a counterpart would be unitary. Real-time versions of replica wormholes have been constructed, but their saddles involve complex metrics [57, 58].

If the boundary encodes the bulk redundantly in a quantum error-correcting code, bulk factorization eventually fails [59, 60, 61] as well.¹ In this picture the states of the bulk EFT on a fixed background form a ‘code subspace’ $\mathcal{H}_{code}$, which a map $V : \mathcal{H}_{code} \rightarrow \mathcal{H}_{CFT}$ embeds into the boundary Hilbert space. A bulk operator ϕ in the entanglement wedge of a boundary region A can be reconstructed on A : there is an operator O_A acting only on A with

$$O_A V|\psi\rangle = V\phi|\psi\rangle \quad \text{for all } |\psi\rangle \in \mathcal{H}_{code} \quad (13)$$

[60]. After the Page time the entanglement wedge of the radiation includes the island, so interior operators can be reconstructed on the radiation [52, 51]. The interior is then not an independent tensor factor alongside the radiation.

A simple count of dimensions shows why this has to happen. The Hilbert space of the

¹A minimal example is the three-qutrit code, $|j\rangle_L \mapsto \frac{1}{\sqrt{3}} \sum_{k=0}^2 |k, k+j, k+2j\rangle \pmod{3}$. Erasing any one physical qutrit leaves the logical state recoverable from the other two [60]. A bulk operator deep in the interior is similarly represented in many different boundary regions.

black hole in the fundamental description has dimension about $e^{S_{BH}}$, which shrinks as the black hole evaporates. The effective description, by contrast, contains an interior partner for every Hawking quantum emitted, so the dimension of its interior Hilbert space grows. After the Page time the effective interior has more states than the fundamental black hole, and the map from effective to fundamental states cannot be an isometry: some effective states must map to the same fundamental state or to zero. Akers, Engelhardt, Harlow, Penington and Vardhan [37] construct such non-isometric maps and show that they still preserve inner products, up to exponentially small errors, for states that can be prepared with sub-exponential complexity. Semiclassical gravity therefore still works for simple observables, even though its Hilbert space is not the Hilbert space of the effective system. With a unitary effective description, the bound of Section 4.2 then requires that interior/exterior factorization eventually fail completely. In other words, while the coherent-state reading of the spacetime background survives for expectation values of simple observables, the state space of the EFT on the background is not the state space of the black hole.

Holography of information [30, 62, 63] takes this further, and its intuition comes from treating gravity as a gauge theory. In electromagnetism, Gauss's law $\nabla \cdot \mathbf{E} = \rho$ means that the total charge inside any region is fixed by the electric flux through its boundary, $Q = \oint \mathbf{E} \cdot d\mathbf{A}$. A charged operator $\psi(x)$ is not gauge invariant on its own. It becomes gauge invariant only with a Wilson line that runs from x to infinity,

$$\psi(x) \exp\left(iq \int_x^\infty A_\mu dx^\mu\right), \quad (14)$$

so creating a charge anywhere inside also changes the field at the boundary. Gravity has the same structure with energy as its charge. The Hamiltonian constraint makes the bulk part of the Hamiltonian vanish on physical states, so the total energy is a boundary integral, the ADM mass,

$$H = \frac{1}{16\pi G_N} \oint_{S_\infty} (\partial_j h_{ij} - \partial_i h_{jj}) dS^i, \quad (15)$$

where h_{ij} is the deviation of the spatial metric from flatness. Any operator that creates an excitation, including one in the black hole interior, must be dressed gravitationally to infinity and changes the energy measured there. The analogy fails in one respect though: in electromagnetism, neutral excitations leave no trace at the boundary, but in gravity every excitation carries positive energy, so nothing is neutral. Raju and collaborators use this to argue that all information about the bulk state is available in the algebra of operators near the boundary. For example, if the theory has a unique vacuum $|\Omega\rangle$ with an energy gap, the projector onto the vacuum is a function of the boundary Hamiltonian, $|\Omega\rangle\langle\Omega| = \lim_{\beta \rightarrow \infty} e^{-\beta(H-E_0)}$, and products of this projector with other boundary operators can be used to distinguish any two bulk states [63, 64].

In this sense (according to holography of information), the black hole exterior already has

the records of the interior and interior/exterior factorization fails from the start. For this reason holography of information does not recover the Page curve at all: the fine-grained entropy of the exterior, defined through the algebra of operators near the boundary, stays constant as the black hole evaporates. For a black hole formed from a pure state in AdS this constant is zero, so the exterior is always pure. In asymptotically flat spacetime the constant need not vanish, because the analysis at null infinity leaves out massive particles [63, 30, 62].

5.4 Ensembles

Holographic models in two dimensions add a further complication. Saad, Shenker and Stanford showed that the JT gravity path integral with one asymptotic boundary, summed over topologies, agrees order by order in the genus expansion with the double-scaled matrix integral

$$\langle Z(\beta) \rangle = \frac{1}{\mathcal{Z}} \int dH e^{-L \text{Tr} V(H)} \text{Tr} e^{-\beta H}, \quad \mathcal{Z} = \int dH e^{-L \text{Tr} V(H)}, \quad (16)$$

over Hermitian $L \times L$ matrices H , with V chosen so that the leading eigenvalue density reproduces the disk (Schwarzian) density of states [65]. The correspondence extends to several boundaries: products such as $\langle Z(\beta_1) Z(\beta_2) \rangle$ require wormholes connecting boundaries that were previously discarded, and these results are then naturally re-interpreted as ensemble averages. If semiclassical gravity describes only an ensemble average, the monotonic growth of its entropy need not describe any actual member, and the Page curve follows from typicality [66].

The problem is that this holography is between two dual *effective* descriptions, not the full quantum gravity theory. We would then be able to pick the actual effective description in the random matrix theory view instead of the JT gravity ensemble view. We then have to ask whether this actual description differs from GR by only small corrections. Therefore, the ensemble picture cannot help here.

We do not discuss the non-ensemble baby-universe interpretations [67, 68, 59] as they matter little for this paper, except that they too are bound by the Page curve on the effective description. We note only that firewalls in JT gravity with replica wormholes raise further complications [69, 70], leading to questions of whether corrections to GR are really small.

5.5 Discussion: what string theory does and does not justify with regards to general relativity

Each string-theoretic response to the information problem goes beyond at least one constraint required for the worldsheet derivation. String spreading exploits the scattering regime. Fuzzballs change the global structure. Replica wormholes are non-perturbative and change topology. Error-correcting codes and holography of information change the state space while leaving expectation values of simple observables intact. In each case the worldsheet derivation still holds

where it applies: the spacetime background obeys the local field equations, and perturbative string states scatter on this background as EFT predicts. This does not fully justify semiclassical gravity as the effective description of an evaporating black hole.

6 Conclusion

String theory derives Einstein's equations from worldsheet conformal invariance, and this is often taken to justify the conventional view that GR, together with EFT on curved spacetime, is an effective description wherever curvature is small. The black hole information problem tests this at the horizon of a large black hole, where the curvature is small in string units.

We noted that the effective description, not only the fine-grained theory, must follow the Page curve, and the Page time falls at low curvature. Small corrections perturbative in the string coupling cannot produce the required Page-curve turnaround, and non-perturbative corrections can produce it only by giving up the no-drama equivalence principle, interior/exterior factorization, or effective unitarity. This necessitates violating the constraints required for the worldsheet derivation: string spreading exploits the scattering regime, fuzzballs change the global structure, replica wormholes are non-perturbative and change topology, and error-correcting codes change the state space.

While the string-theoretic GR derivation remains correct for local field equations and perturbative scattering at fixed topology, it does not fully justify semiclassical gravity as the effective description of an evaporating black hole, and mainstream approaches to the black hole information problem fully exploit this open gap.

Despite the optimism of the mainstream approaches, at present there is no evidence that GR breaks down near black hole horizons, and it is unclear how, on the mainstream approaches, the failure of semiclassical gravity could stay invisible if it must. In light of this, Schneider's argument [4, 47] that the information problem should be resolved by rethinking global spacetime structure, rather than by modifying the local content of the effective theory, may well be correct. For now, neither possibility can be ruled out. What we claim in this paper is only that the string-theoretic support for semiclassical gravity as an effective description is weaker than often thought.

Data availability and declaration of interests

The author(s) have no funding source to declare. There is no conflict of interests.

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